

Answers to Quiz 7: Rectangle and Trapezoidal Rules

(Numerical Analysis)

1. Approximate values needed for the plot are given in the table below.

Note: you don't need to calculate $f(x_i^*)$. It is given for reference only.

i	0	1	2	3	4	5
x_i	-1.00	-0.21	0.46	1.03	1.54	2.00
x_i^*	—	-0.37	0.32	0.92	1.44	1.91
$f(x_i^*)$	—	-0.22	-0.097	-0.28	0.40	2.40

Table 1: Approximate values of x_i , x_i^* , and $f(x_i^*)$

- 2.

$$I = \int_0^{3\pi/2} \sin(x) dx = -\cos(x) \Big|_0^{3\pi/2} = 1$$

$$\tilde{I} = -\frac{\pi}{8} + \frac{\pi}{4} \left(1 + \frac{1}{\sqrt{2}} \right) \approx 0.948$$

$$e_a \approx |1 - 0.948| \approx 0.052$$